

Polimaly: Industrial Anomaly Detection in Kaputt2 Dataset Retail Track – Off-the-shelf VLM Setting

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Abstract

Visual anomaly detection in retail environments remains challenging due to large object diversity, unconstrained viewpoints, and subtle defect patterns. In this work, we present a zero-shot anomaly detection approach for the VAND 4.0 Retail Track on the Kaputt2 dataset using off-the-shelf Vision-Language Models (VLMs). Our method leverages the multimodal reasoning capabilities of Qwen3.5-9B without any fine-tuning or parameter optimization. We formulate anomaly detection as a binary yes/no classification task guided by a handcrafted prompt describing structural and logical defect categories. Instead of relying on free-form textual outputs, we compute anomaly scores directly from the logits of the generated answer tokens, leading to more stable and discriminative predictions. Through extensive prompt engineering, we show that concise prompts generalize better across diverse retail products than longer product-specific descriptions. Experimental results demonstrate that our approach significantly outperforms previously reported off-the-shelf VLM baselines on the Kaputt benchmark, achieving 52.71% AP_{any} and 68.05 AUROC in the no-training, no-reference setting. Our findings highlight the potential of large pretrained VLMs for scalable and adaptable retail anomaly detection under realistic zero-shot conditions.

The implementation code will be publicly available by May 17 on this Github repository.¹

1. Introduction

1.1. Background

Automated visual anomaly detection plays a critical role in industrial quality assurance and retail logistics, where defective products must be identified efficiently and reli-

ably before reaching customers. While recent anomaly detection benchmarks such as MVTec-AD [3] and VisA [9] have enabled major progress in industrial inspection, these datasets mostly focus on controlled manufacturing environments with limited object diversity and stable viewpoints. In contrast, retail logistics introduces substantially more challenging conditions, including large intra-class variation, unconstrained object poses, subtle defects, and limited availability of reference samples.

The Kaputt dataset [4] was introduced to address these limitations by providing a large-scale benchmark tailored to real-world retail defect detection. The dataset contains more than 238k images covering over 48k unique products and includes 7 different defect categories. Unlike traditional industrial anomaly detection settings, Kaputt emphasizes scenarios where products appear in different poses, packaging variations, and visually ambiguous conditions, making generalization significantly harder for existing anomaly detection methods. Previous evaluations on Kaputt demonstrated that state-of-the-art unsupervised anomaly detection and vision-language models (VLMs) struggle under these conditions, often performing only slightly above chance level.

These challenges make retail anomaly detection highly relevant for modern logistics systems, where scalable and adaptable solutions are required to handle continuously changing inventories without expensive annotation or re-training pipelines. In this context, off-the-shelf VLMs provide an attractive direction due to their strong visual reasoning capabilities and zero-shot generalization potential.

1.2. Challenge Description

This work was developed for the VAND 4.0 Retail Track, which evaluates visual defect detection methods on the Kaputt2 dataset, a test-only extension of the original Kaputt benchmark. The challenge focuses on detecting defective retail items under realistic logistics conditions characterized by strong pose variation, diverse product categories, and subtle visual anomalies.

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¹<https://github.com/Amirparsa-Sal/Polimaly-Anomaly-Detection>

We participate in the Off-the-Shelf VLM setting, where participants are restricted to using pre-trained vision-language models without any form of weight modification, fine-tuning, or parameter optimization. The setting supports two evaluation scenarios: (1) zero-shot inference without reference images, and (2) few-shot inference using up to three reference images per item during test time. Training on Kaputt2 data is strictly prohibited, while prompt engineering, image preprocessing, and in-context learning strategies are permitted.

The primary evaluation metric of the challenge is the per-class Average Precision on any defect (AP_{any}), which measures the ability of a method to distinguish defective from non-defective products across varying confidence thresholds.

2. Methodology

2.1. Model Design

Our approach focuses on zero-shot anomaly detection using off-the-shelf Vision-Language Models (VLMs). We explored different prompting and inference strategies for both zero-shot and few-shot settings, with the goal of leveraging the reasoning capabilities of large multimodal models without any task-specific training or fine-tuning.

2.1.1. Approach

During our experiments, we evaluated multiple models from the Qwen family [8] for both zero-shot and few-shot anomaly detection. While several approaches using reference images were explored, we observed that the zero-shot setting consistently achieved better performance and more stable predictions.

Our final method uses a single prompt describing different categories of anomalies and instructs the model to answer with either `yes` or `no` depending on whether the item is defective. Instead of relying on free-form textual outputs or directly asking the model for a numerical defect score, we compute the final anomaly score using the logits corresponding to the generated answer tokens.

Specifically, we extract the logits for different capitalization variants of the tokens “yes” and “no”:

$$\text{logits}_{yes} = \max(\text{logit}(\text{yes}), \text{logit}(\text{YES}), \text{logit}(\text{Yes}))$$

$$\text{logits}_{no} = \max(\text{logit}(\text{no}), \text{logit}(\text{NO}), \text{logit}(\text{No}))$$

The final anomaly score is then computed using a softmax formulation:

$$\text{score} = \frac{e^{\text{logits}_{yes}}}{e^{\text{logits}_{yes}} + e^{\text{logits}_{no}}}$$

Through empirical analysis, we found that this binary logit-based scoring strategy produces more consistent predictions than directly requesting a scalar anomaly score from the model. We hypothesize that directly predicting numerical scores requires the model to maintain a relative notion of anomaly severity across samples, which is difficult in a stateless zero-shot setting.

2.1.2. Architecture

Our final model is based on Qwen3.5-9B, a large multimodal Vision-Language Model designed for visual understanding and instruction-following tasks. The Qwen family provides strong zero-shot reasoning capabilities and supports image-text alignment through a unified multimodal architecture. Compared to other variants, the 9B model demonstrated better visual reasoning performance and more reliable anomaly detection behavior in our experiments, while still remaining computationally feasible for large-scale inference.

2.1.3. Prompting

Our final prompt is shown in Figure 1. The prompt organizes anomalies into two high-level categories: *structural anomalies* and *logical anomalies*. For each category, we describe a set of visual indicators that may suggest a defective product. Structural anomalies mainly correspond to physical defects such as deformation, damage, or missing parts, while logical anomalies refer to inconsistencies in the arrangement, count, orientation, or composition of product components.

The prompt was developed through an iterative prompt engineering and optimization process. To design it, we first analyzed the training set and studied the provided defect labels to better understand the different anomaly types present in the dataset. We then performed a visual inspection of representative samples from each defect category in order to identify common patterns and recurring visual cues associated with defective products. From this analysis, we extracted multiple defect scenarios and translated them into concise textual descriptions that could guide the vision-language model during inference.

We also investigate several prompt formulation strategies. One alternative approach grouped anomalies according to product categories, such as boxes, containers, wrapped items, and multi-part products. However, we observed that shorter and more direct prompts generalized better across diverse object types and produced more stable predictions. Consequently, the final prompt focuses on concise descriptions of anomaly characteristics rather than detailed product-specific instructions.

2.2. Dataset & Evaluation

2.2.1. Data Utilization

We did not apply any explicit data augmentation or image pre-processing techniques during inference. Instead, we directly used the cropped item images provided as part of the Kaputt dataset.

As discussed in the Kaputt benchmark paper, model performance can vary significantly depending on the image representation used, such as full tray images, masked images, or cropped item regions [4]. During our experiments, we observed that the cropped images consistently produced better results for our VLM-based approach.

We believe this behavior is mainly due to the higher effective resolution of the object region in the cropped images, which allows the model to focus more effectively on fine-grained visual details and subtle anomalies.

2.2.2. Evaluation Criteria

The challenge primarily evaluates methods using two metrics:

- **Average Precision on any defect (AP_{any}):** evaluates performance on detecting both *minor* and *major* defects. It is computed as the area under the Precision–Recall curve:

$$AP = \sum_n (R_n - R_{n-1}) P_n$$

where P_n and R_n denote precision and recall at threshold n .

- **Area Under the Receiver Operating Characteristic Curve (AUROC):** measures the trade-off between the True Positive Rate (TPR) and False Positive Rate (FPR) across different thresholds:

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

Higher AUROC values indicate better separation between defective and non-defective samples.

Although additional metrics are reported for analysis, the official challenge ranking is determined solely by AP_{any} .

2.2.3. Evaluation Scenario

The Kaputt benchmark defines four evaluation scenarios depending on whether training data and reference images are available during inference:

- **No training, no reference:** zero-shot methods that perform defect detection without task-specific training or reference images.

Zero-Shot Quality Inspection Prompt

System: You are a quality inspector. Answer questions with only ‘Yes’ or ‘No’. Do not explain or think step by step.

User: Examine this product image carefully for any signs of damage or defect.

Consider the following **structural** defect types:

- **Penetration:** holes, tears, or cuts in packaging or product
- **Deformation:** dents, crushes, bent or warped surfaces, rounded or ruined corners
- **Deconstruction:** detached, collapsed, or disassembled parts
- **Spillage:** leaked or spilled contents (liquid, powder, granules)
- **Superficial:** surface damage such as dirt marks, spots, stains, scratches, or scuffs

And also the following **logical** defect types:

- **Actuation:** inappropriately opened containers (open box, bag, or book), or not properly wrapped in plastic objects.
- **Missing Unit:** one or more expected parts or units are absent

Is there any defect visible in this image? Answer only **Yes** or **No**.

Figure 1. Zero-shot prompt template used for defect detection.

- **No training, with reference:** few-shot anomaly detection methods that use reference images at inference time without additional training.
- **With training, no reference:** supervised methods trained on labeled defect data without using reference images.
- **With training, with reference:** methods that combine supervised training with reference-image-based reasoning.

Since we use a pre-trained VLM, our work focuses exclusively on the *no training* scenarios. During development, we explored multiple strategies for few-shot detection using reference images. However, our final approach is a simple zero-shot method that does not use reference images during inference. Therefore, our submission belongs to the **no training, no reference** scenario.

3. Results

3.1. Performance Metrics

We evaluate our model primarily using the metrics described in Section 2.2.2, namely AP_{any} and AUROC. In addition, we use box plots to analyze the prediction distribution for each defect category individually in order to bet-

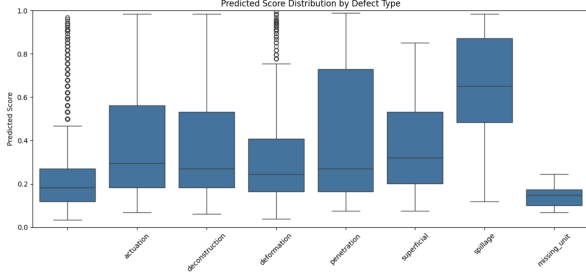


Figure 2. The per-class box-plot of the predictions of the model

ter understand the strengths and weaknesses of the model.

Table 2 presents the performance of different variants of the Qwen family on a subset of the Kaputt1 training set. Due to computational and time limitations, we were not able to evaluate all models on the complete dataset. While Qwen3.6-27B achieved the highest AUROC, the improvement over Qwen3.5-9B was relatively small. In contrast, Qwen3.5-9B achieved the best AP_{any} score while requiring substantially fewer computational resources. Considering this performance–efficiency trade-off, we selected Qwen3.5-9B as the final architecture for all subsequent experiments. Notably, both 27B models were evaluated using FP8 precision to reduce memory and computational requirements.

Figure 2 illustrates the distribution of model predictions for the samples from the test set of Kaputt1 dataset, grouped by their defect type. As shown in the figure, the model assigns very high anomaly scores to the *spillage* class, indicating strong capability in detecting content leakage and related anomalies. The model also achieves relatively strong performance on *superficial*, *deconstruction*, *penetration*, and *actuation* defects. Although the mean predictions for these classes are generally higher than for normal samples, the prediction ranges remain wide, suggesting inconsistent behavior across samples.

The main failure cases are the *deformation* and *missing unit* categories. Deformation defects are often subtle and localized around edges or corners, making them difficult to distinguish from normal appearance variations or lighting artifacts. The poor performance on missing unit defects is likely caused by two factors: the absence of reference-image reasoning and the use of cropped images, which may exclude detached or missing object components from the visible region.

Finally, the model occasionally produces high anomaly scores for non-defective samples, which negatively affects overall performance. Moreover, the mean prediction values for normal samples are relatively close to those of the deformation and missing unit categories, further confirming that these two defect types remain challenging for the model.

Baseline	Tray/Item	AP_{any} [%]	AUROC
<i>Random</i>	-	31.84	50.00
CLIP [6]	item	36.20	56.05
POMP [7]	item	32.98	50.44
WinCLIP-zero [5]	item	33.87	52.30
Claude-icl [2]	tray	36.57	56.96
Pixtral-zero [1]	tray	32.75	50.93
Pixtral-icl	tray	32.18	50.86
Ours	item	52.71	68.05

Table 1. Performance comparison for the no-training, no-reference setting, tested on the Kaputt1 test set split with 10067 total samples including 3206 defective samples

Model	Tray/Item	AP_{any} [%]	AUROC
<i>Random</i>	-	29.00	50.00
Qwen3.5-4B	item	40.42	60.50
Qwen3.5-9B	item	47.11	66.64
Qwen3.5-27B	item	43.43	64.29
Qwen3.6-27B	item	46.43	67.17

Table 2. Performance of different Qwen variants on a subsample of the Kaputt1 training images with 3000 total samples including 870 defective samples

3.2. Comparison

Table 1 compares our final model against the baseline methods reported in the Kaputt benchmark paper. Our approach achieves an AP_{any} score of 52.71% and an AUROC of 68.05, substantially outperforming previously reported off-the-shelf VLM baselines.

We believe this improvement mainly stems from two factors: the strong multimodal reasoning capabilities of the Qwen3.5 model and the extensive prompt engineering process used during development. In particular, the carefully designed binary prompting strategy and logit-based scoring mechanism contributed significantly to obtaining more stable and discriminative anomaly predictions.

4. Discussion

4.1. Challenges & Solutions

One of the main challenges of the Kaputt benchmark is the large variety of anomaly types. Since the model must detect multiple categories of defects with different visual characteristics, designing prompts that comprehensively describe all anomaly types becomes difficult. Longer prompts, however, often reduce the effectiveness of the model by diluting important instructions and causing the VLM to lose focus on fine-grained visual details in the image. To address this

issue, we iteratively optimized our prompts and observed that shorter and more direct prompts generally produced more stable predictions.

Another challenge is the extreme diversity of target objects in the dataset. Unlike traditional industrial anomaly detection benchmarks that focus on a small set of products, Kaputt contains thousands of visually distinct items. This motivated our use of Vision-Language Models, since these models are pretrained on large-scale multimodal datasets and therefore possess strong zero-shot generalization capabilities across diverse object categories.

Certain anomaly types are inherently more difficult to detect than others. In particular, deformation defects are often subtle and localized around edges or corners of the object, making them difficult to distinguish from natural shape variations, lighting effects, or perspective distortions. This challenge is reflected in the lower prediction confidence observed for this category.

We also explored few-shot approaches using reference images. However, an important limitation of the benchmark is that not all reference images are guaranteed to be defect-free. As noted in the Kaputt paper, some reference images may themselves contain anomalies or packaging variations. This makes it difficult for the model to reliably infer a notion of normality from the reference set and often introduces additional ambiguity during inference.

4.2. Model Robustness & Adaptability

A major advantage of our approach is its strong zero-shot generalization capability. Since the model is not trained or fine-tuned on the Kaputt dataset, it remains highly adaptable to previously unseen products and anomaly types. This property is especially important in retail logistics environments, where the inventory changes continuously and collecting labeled training data for every new product is impractical.

By leveraging a large pretrained VLM, our method benefits from broad visual knowledge acquired during pretraining on diverse image-text datasets. As a result, the model can reason about a wide range of objects and defect patterns without requiring task-specific optimization. This makes the approach naturally scalable to real-world anomaly detection scenarios with large object diversity.

4.3. Off-the-Shelf VLM / Zero-Shot Setting

Our method belongs to the *Off-the-Shelf VLM* setting defined by the challenge. The model weights remain completely unchanged throughout development and evaluation, and no fine-tuning, adapter training, or parameter optimization is performed.

Furthermore, our final approach operates in a fully zero-shot manner. During inference, the model receives only the query image together with a handcrafted prompt describ-

ing possible anomalies, without using any training samples or reference images. The anomaly score is then computed directly from the model’s output logits corresponding to binary yes/no predictions.

4.4. Future Work

Although we explored several prompting strategies for comparing query images against reference images, we were unable to achieve better performance than the pure zero-shot setting. A promising direction for future work would be designing more effective comparative prompting methods that allow the model to refine its predictions using reference-image information. However, such approaches would also require robust mechanisms for identifying reliable and non-defective reference images within the reference set.

Another potential direction is incorporating image pre-processing or enhancement techniques into the pipeline. Our current approach uses the provided cropped images directly without any additional preprocessing. Techniques such as adaptive cropping, lighting normalization, or multi-scale analysis may further improve the model’s ability to detect subtle anomalies.

5. Conclusion

In this work, we presented a zero-shot retail anomaly detection method based on an off-the-shelf Vision-Language Model for the VAND 4.0 Retail Track on the Kaputt2 dataset. Our approach uses Qwen3.5-9B together with a carefully designed binary prompting strategy and a logit-based anomaly scoring mechanism, without any fine-tuning or task-specific training. The proposed method achieved strong performance compared to previously reported off-the-shelf VLM baselines, demonstrating the effectiveness of large multimodal models for retail defect detection under realistic conditions.

Our experiments showed that concise and general anomaly descriptions lead to more stable predictions than long product-specific prompts. We also observed that directly using binary answer logits provides more reliable anomaly scores than requesting scalar confidence values from the model. While the approach performs well on several defect categories such as *spillage* and *superficial* damage, some anomalies like *deformation* and *missing units* remain challenging due to the complexity of fine-grained visual differences and the absence of reference reasoning, respectively.

Overall, the results confirm that pretrained VLMs possess strong zero-shot generalization capabilities for large-scale retail inspection tasks involving diverse product categories. We believe that future improvements in comparative prompting, reference-image selection, and image pre-processing could further enhance the robustness and accuracy of zero-shot anomaly detection systems in real-world

logistics environments.

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